

Public Goods and Common Resources

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The core text is N. Gregory Mankiw, *Principles of Microeconomics*.

- **Chapter 11** — Public Goods and Common Resources.

Read the chapter alongside these slides.

Where we are going

1. **Two questions to ask of any good** — the map: four kinds of goods.
2. **The free-rider problem** — why the hat comes back light.
3. **Should the government build it?** — cost–benefit analysis, and a vote that goes wrong.
4. **The tragedy of the commons** — everybody's property is nobody's.
5. **The commons, with numbers** — the overuse, counted exactly.
6. **Fixing the commons** — cap it, price it, permit it, own it.

PART 1

Two Questions to Ask of Any Good

Goods without price tags

Some goods reach you through a checkout: a burrito, a haircut, a tank of gas.

Others reach you with no transaction at all: street lighting, clean harbor water, national defense, the road you drove in on.

Prices are how markets ration goods and reward suppliers.

Where no price can be charged, both jobs go undone. So: **when** does that happen, and **what** happens next?

Question one: can you be kept out?

Excludable. A good is excludable if a supplier can prevent people who have not paid from using it.

The test: **is there a gate?** Can non-payers be kept out at reasonable cost?

- **Excludable:** a burrito (hand it over or not); a streaming subscription (password).
- **Not excludable:** street lighting; fish swimming in the open ocean; national defense.

Question two: does your use leave less?

Rival in use. A good is rival if one person's use of a unit reduces what is left for everyone else.

The test: **if I use it, is there less for you?**

- **Rival:** a burrito (eaten once); a fish (caught once); a parking space at 9 a.m.
- **Not rival:** a streamed show (your watching takes nothing from mine); street lighting; national defense.

Active Learning 1

For each good, answer the two questions — **excludable?** and **rival?** — with yes or no:

1. a burrito
2. a streaming subscription
3. fish in the open ocean
4. national defense
5. street lighting
6. a crowded free beach

Two minutes. Paper and pencil.

Two questions, four kinds of good

		Rival in use?	
		yes	no
Excludable?	yes	?	?
	no	?	?

Every good you will ever meet lands in exactly one cell — **and the cell predicts whether a market can handle it.**

The gated row

		Rival in use?	
		yes	no
Excludable?	yes	Private goods a burrito	Club goods a streaming subscription
	no	?	?

Where there is a gate, a price can be charged — suppliers get paid, and markets form.

Club goods: the gate works even though sharing costs nothing, so access is sold by subscription.

The open row

		Rival in use?	
		yes	no
Excludable?	yes	Private goods a burrito	Club goods a streaming subscription
	no	Common resources fish in the open ocean	Public goods national defense

No gate, no price — the bottom row is where markets fail.

Goods move cells

A city street, under four conditions:

	excludable?	rival?	cell
free, empty	no	no	public good
free, jammed	no	yes	common resource
tolled, empty	yes	no	club good
tolled, jammed	yes	yes	private good

Congestion switches rivalry on. Tolling technology switches excludability on.

The cell is not a fixed property of the good.

Active Learning 2

The campus gym:

1. Members-only, at 2 p.m. (nearly empty) — which cell?
2. Members-only, at 6 p.m. (every machine taken) — which cell?
3. Open-to-everyone rec room, 2 p.m. — which cell?
4. Open-to-everyone rec room, 6 p.m. — which cell?
5. **The why is the marks:** name one good that has moved cells *in your lifetime* because technology changed, and say which question flipped.

Three minutes. Paper and pencil.

PART 2

The Free-Rider Problem

Start in the bottom-right cell

		Rival in use?	
		yes	no
Excludable?	yes	Private goods a burrito	Club goods a streaming subscription
	no	Common resources fish in the open ocean	Public goods national defense

Not excludable, not rival: nobody can be kept out, and nobody's use leaves less. **Who would ever supply this for a living?**

A neighborhood of **120 households** around a pond. Mosquitoes, every summer.

A treatment program knocks them down for the season. Cost: **\$2,500 per season**.

Run the two questions:

- Excludable? **No** — fewer mosquitoes over the pond means fewer for every household; no gate can be built.
- Rival? **No** — one family's itch-free evening takes nothing from the next family's.

A public good, worth real money to real people.

The neighborhood association asks every household for **\$25** toward the program ($120 \times \$25 = \$3,000 \geq \$2,500$).

Free rider. Someone who enjoys the benefit of a good without paying for it.

The program cannot exclude non-payers. So before writing a check, each household can ask: *what does my \$25 actually change?*

The free rider's arithmetic

Take a household that values the season at **\$50**. With 120 households, its \$25 **does not tip** whether the program runs.

	program runs anyway	program fails anyway
chip in \$25	$50 - 25 = \$25$	-\$25
keep it	\$50	\$0

Keeping the \$25 is better in both columns.

The hat comes back light

Every household faces the same two columns — **including the ones that value the program most.**

So the hat comes back with a fraction of \$2,500, and the program dies — *even though the neighborhood values it at well above its cost.*

Nobody cheated. Everybody reasoned. The good still died.

No gate, no revenue, no supply

Why the market fails here, in one chain:

no exclusion $\Rightarrow$ willingness to pay never becomes revenue $\Rightarrow$ **no private supplier can cover cost**

The demand is real — \$50 households exist. The failure is that **no price can reach it.**

Public goods are **undersupplied** by markets — often supplied not at all. This is market failure by *omission*.

Active Learning 3

1. A household values the season at \$50, is asked for \$25, and keeps its money.
Using the two-column table, say **why** this is rational, not selfishness gone wrong.
2. A local hardware store offers to **match every dollar** collected, one-for-one.
Predict, then check with the table: does matching fix the free-rider problem?
3. Would the hat work better on a **cul-de-sac of 5 households**? What is different?
The why is the marks.

Four minutes. Paper and pencil.

PART 3

Should the Government Build It?

What taxing can do that asking cannot

A town government holds the one power the neighborhood association lacks: **it can make paying compulsory.**

Tax every household, hire the crew — the free rider's two columns vanish, because *keep it* is no longer on the menu.

But compulsion cuts both ways. Which projects deserve it?

Cost–benefit analysis. Comparing the *total* benefit of a project to everyone with its *total* cost, to decide whether it is worth doing.

Write the test

Build the program if total benefit $>$ total cost.

Total benefit = each household's value, summed over all 120:

$$B = 40 \times \$50 + 60 \times \$20 + 20 \times \$5$$

Read the pieces:

- 40 pond-front households would pay up to \$50 each,
- 60 set back from the water, up to \$20 each,
- 20 on the far ridge, barely touched, up to \$5 each.

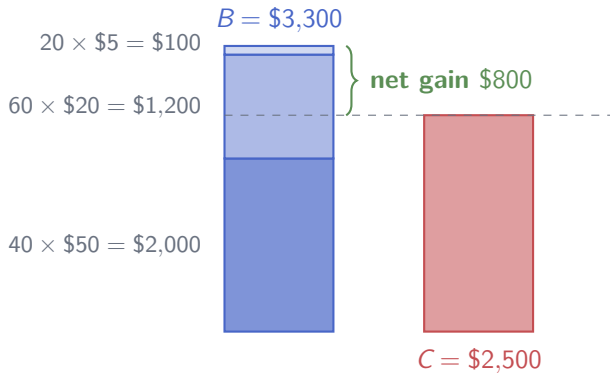
Cost: $C = \$2,500$.

Plug the numbers in

group	benefit	running total
40 households $\times$ \$50	$= \$2,000$	\$2,000
60 households $\times$ \$20	$= \$1,200$	\$3,200
20 households $\times$ \$5	$= \$100$	\$3,300
total benefit B		\$3,300
total cost C		\$2,500
net gain $B - C$		$3,300 - 2,500 = \$800$

Verdict: build it. The town is \$800 per season richer with the program than without it.

Draw the comparison



The benefit column, stacked by group, clears the cost line by **\$800**.

Where would the numbers come from?

The test needed each household's true value. But a public good has **no market and no price** — there is nothing to observe.

So the town must *ask*. And asking invites strategy:

- If your stated value sets your **tax** → understate it.
- If it just makes the project likelier → **overstate** it.

The self-interest that emptied the hat also bends the survey.

In practice: benefits are *estimates* — cost–benefit analysis is honest arithmetic on uncertain inputs.

Put it to a vote

Fund it the obvious way: **equal shares**, $2,500/120 \approx \$20.83$ per household.

Referendum:

group	value	tax	net	vote
40 households	\$50	\$20.83	$50 - 20.83 = +\$29.17$	yes
60 households	\$20	\$20.83	$20 - 20.83 = -\$0.83$	no
20 households	\$5	\$20.83	$5 - 20.83 = -\$15.83$	no
result	no wins, 80–40			

An efficient project loses two to one.

Change the split, change the vote

Same project, same \$2,500 — **shaped to benefit** instead of split equally:

group	value	tax	revenue	net
40 households	\$50	\$38	$40 \times 38 = \$1,520$	+\$12 each
60 households	\$20	\$15	$60 \times 15 = \$900$	+\$5 each
20 households	\$5	\$4	$20 \times 4 = \$80$	+\$1 each
total			\$2,500	yes, 120–0

The obstacle was the split, not the project. But shaping taxes to benefit requires knowing the values — *the very numbers people hide*.

National defense — the classic pure case: roughly \$900 billion of U.S. spending in 2024, covering every resident, excluding none.

Basic research — an idea, once published, is non-rival and non-excludable: a theorem protects no one who funds it.

- A **patent** builds a legal gate around an *applied* idea → moves invention up to the club/private row → private R&D happens.
- Basic science has no such gate — so it runs on public funding and philanthropy.

The cell describes the good, not the supplier

public goods, privately supplied	private goods, publicly supplied
Wikipedia (donations, volunteers)	a public-school seat
ad-funded broadcast radio	(rival <i>and</i> excludable)
open-source software	a city bus ride

Public good is a statement about **excludability and rivalry** — never about who happens to provide it.

Private provision of public goods is real but fragile: it leans on donations, advertising, or volunteers — **not on a price for the good itself**.

Active Learning 4

The pond committee proposes a **second** treatment round in August. Cost: \$1,800. Honest values for the extra round: 40 households at \$20, 60 at \$10, 20 at \$0.

1. **Predict before calculating:** values fell but so did the cost — can you tell the verdict without arithmetic?
2. Run the test. Build or not?
3. **Check it a second way**, per household: compare *average* benefit with *average* cost. Why must the two routes agree?
4. A no-taxes-attached survey reports values of 40 at \$40, 60 at \$20, 20 at \$5. Recompute. **Which total do you believe, and why? The why is the marks.**

Five minutes. Paper and pencil.

PART 4

The Tragedy of the Commons

The dangerous cell

		Rival in use?	
		yes	no
Excludable?	yes	Private goods a burrito	Club goods a streaming subscription
	no	Common resources fish in the open ocean	Public goods national defense

Common resource. A good that is rival in use but not excludable: anyone may take, and every taking leaves less.

An old story

A village pasture, open to every herder.

- Each animal added is **pure gain to its owner** — the grass it eats is free.
- The grass it eats is **loss to every herd on the pasture** — shared by all, charged to none.
- So herds grow past what the pasture can renew... and the pasture dies. Everyone's animals starve.

The name comes from Garrett Hardin's 1968 essay "The Tragedy of the Commons" (*Science*).

The diagnosis: a cost nobody pays

When my use of a good imposes a cost on you that I never pay, **I use too much of it.**
That is the whole mechanism.

The herder counts *all* of the added animal's value — but bears only a *sliver* of the grass it destroys.

Public goods: too little gets made.

Common resources: too much gets taken.

Two cells, one missing gate — failing in opposite directions.

- **Ocean fisheries** — the northern cod off Newfoundland collapsed; Canada closed the fishery in 1992, and it has never fully returned.
- **Groundwater** — the Ogallala aquifer, drawn down faster than rain refills it, well by private well.
- **Rush-hour roads** — every driver slows every other driver, and no one is billed for it.
- **Antibiotics** — each course of use erodes a shared pool of effectiveness that no prescriber pays for.

No villains required

The tragedy needs **no greed, no malice, no fools** — only ordinary self-interest plus open access.

Blaming the users misreads the mechanism: *anyone* in their position faces the same arithmetic.

So the repair is never “find better people.”
It is: change the arithmetic.

Active Learning 5

The commuter lot nearest the classroom buildings: **free, unassigned** — and full by 9 a.m. every day.

1. Run the two questions. Which cell is this lot in — and does the cell *change* at 7 a.m.?
2. When one more car takes a space at 8:55, **who bears the cost**, and what is it?
3. **Predict:** the university doubles the lot. What does it look like at 9 a.m. a year later, and why?
4. Name the *smallest* change that would leave ten spaces open at 9 a.m. **The why is the marks.**

Three minutes. Paper and pencil.

PART 5

The Commons, with Numbers

Birch Lake: write the equation

A lake with good fishing, open to all. A boat-day costs **\$400** (fuel, gear, wages).
Boats crowd each other:

$$\text{catch per boat} = 1200 - 100b \quad (b = \text{boats on the water})$$

Read the coefficients:

- **1200**: what one boat alone lands in a day (\$ of fish).
- **100**: what every boat loses when one more launches — the crowding term.

Total catch = boats $\times$ catch per boat: $T(b) = b(1200 - 100b)$.

Plug the numbers in

boats b	catch per boat ($1200 - 100b$)	total catch ($b \times$ per boat)	extra from last boat
1	$1200 - 100 = 1100$	$1 \times 1100 = 1100$	—
2	$1200 - 200 = 1000$	$2 \times 1000 = 2000$	$2000 - 1100 = 900$
3	$1200 - 300 = 900$	$3 \times 900 = 2700$	$2700 - 2000 = 700$
4	$1200 - 400 = 800$	$4 \times 800 = 3200$	$3200 - 2700 = 500$
5	$1200 - 500 = 700$	$5 \times 700 = 3500$	$3500 - 3200 = 300$
6	$1200 - 600 = 600$	$6 \times 600 = 3600$	$3600 - 3500 = 100$
7	$1200 - 700 = 500$	$7 \times 500 = 3500$	$3500 - 3600 = -100$
8	$1200 - 800 = 400$	$8 \times 400 = 3200$	$3200 - 3500 = -300$
9	$1200 - 900 = 300$	$9 \times 300 = 2700$	$2700 - 3200 = -500$

Two columns to watch: what a boat **catches** (per boat), and what a boat **adds** (extra). They are not the same number.

Route one: the net column

boats b	total catch	cost ($400b$)	net (catch – cost)
1	1100	400	$1100 - 400 = 700$
2	2000	800	$2000 - 800 = 1200$
3	2700	1200	$2700 - 1200 = 1500$
4	3200	1600	$3200 - 1600 = 1600$
5	3500	2000	$3500 - 2000 = 1500$
6	3600	2400	$3600 - 2400 = 1200$
7	3500	2800	$3500 - 2800 = 700$
8	3200	3200	$3200 - 3200 = 0$
9	2700	3600	$2700 - 3600 = -900$

The net column peaks at **4 boats: \$1,600 a day.**

Route two: stop when the extra stops covering

Add a boat only while the **extra catch** covers the **\$400** it costs to launch:

boat	extra catch	vs. \$400	verdict
3rd	\$700	$700 > 400$	launch
4th	\$500	$500 > 400$	launch
5th	\$300	$300 < 400$	stay ashore

Stop at **4 boats** — the same answer, and *it must be*: the net column climbs exactly as long as each boat adds more than it costs.

Two routes, one answer: the check that catches errors.

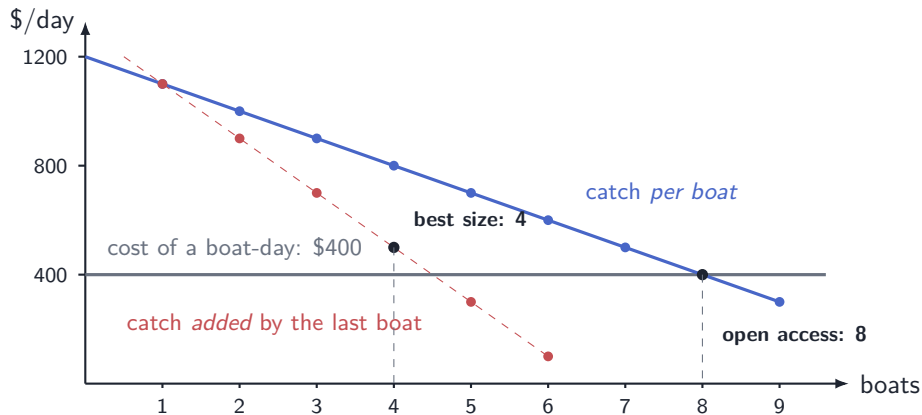
What open access actually does

Nobody asks what their boat *adds*. Each skipper asks: “**will my boat’s catch cover my \$400?**” — the *per-boat* column:

boat	catch per boat	vs. \$400	decision
6th	\$600	$600 > 400$	launch
7th	\$500	$500 > 400$	launch
8th	\$400	$400 = 400$	launch (breaks even)
9th	\$300	$300 < 400$	stays ashore

Entry stops only at **8 boats** — twice the number the lake can profitably carry.

Draw the graph



The lake's rent, eaten alive

	best size (4 boats)	open access (8 boats)
total catch	\$3,200	\$3,200
cost of boats	$4 \times 400 = \$1,600$	$8 \times 400 = \$3,200$
net, per day	\$1,600	\$0

Eight boats burn \$3,200 to land what four boats land for \$1,600.

Open access destroys every dollar the lake was worth.

Each boat exactly breaks even — the lake's *rent* is fully dissipated.

Catch is not contribution

Why does the 5th skipper launch, when the lake wants 4?

the skipper's ledger

my catch ($b = 5$)	+\$700
my cost	-\$400
my profit	+\$300 $\Rightarrow$ launch

the lake's ledger

taken off the 4 boats out (4×100)	-\$400
genuinely new fish ($700 - 400$)	+\$300
cost of the 5th boat	-\$400
the lake's change	-\$100

Of the \$700 caught, \$400 was merely *moved* — taken from other decks.

Active Learning 6

Fuel prices spike: a boat-day now costs **\$600**. Same lake, same catch table.

1. **Predict before calculating:** does the open-access fleet shrink, grow, or stay?
Does the lake's best size change? Does the *gap* survive?
2. Open access: how many boats now? (Which column, which rule?)
3. Best size: how many boats now? (Which column, which rule?)
4. **Check the best size a second way** with a net column at \$600. Why must the routes agree?

Five minutes. Paper and pencil.

PART 6

Fixing the Commons

One problem, four fixes

The broken arithmetic, in one line: **takers never pay the cost their taking imposes on the other users.**

Every working repair re-attaches that cost. Four families:

1. **Cap it** limit use directly
2. **Price it** charge for each use
3. **Permit it** cap, then let the caps trade
4. **Own it** give someone the whole

Each one gets tested on **Birch Lake** — does it get us back to 4 boats and \$1,600 a day?

License exactly 4 boats. Fleet: 4. Net: back to \$1,600 a day. **It works.**

What it demands:

- The regulator must **know the lake's table** — cap at 6 and most of the waste survives; cap at 2 and value is thrown away.
- Someone must **police** the water.
- Someone must decide **who** gets the 4 licenses — queue? lottery? connections?

The real world's version: fishing seasons, catch limits, hunting licenses, well-drilling permits.

Price it

Charge a fee per boat-day — say \$350. A boat-day now costs $400 + 350 = \$750$ *to the skipper*.

boat	catch per boat	vs. \$750	decision
4th	\$800	$800 > 750$	launch
5th	\$700	$700 < 750$	stays ashore

Entry stops at **4 boats** — the fee makes each skipper *feel* the crowding cost they used to dump on the fleet.

Any fee over \$300, up to \$400, lands on the same 4 boats.

Price it: where the value goes

With 4 boats and the \$350 fee, the \$1,600 a day is **preserved** — and split:

- boats keep $4 \times (800 - 750) = \$200$,
- the town collects $4 \times 350 = \$1,400$ — available to cut other taxes.

Congestion pricing is this fix on asphalt.

The IGM panel of leading economists, on peak-hour road pricing: tolls at peak times, with the proceeds used to lower other taxes, “would make citizens on average better off.”

Cap at 4 boat-days — then let the permits trade.

What is a permit worth? Fishing with one clears $800 - 400 = \$400$ a day — so permits sell for **about \$400**, and end up with *the skippers who fish best*.

The cap sets **how much**; the trading decides **who** — the regulator no longer has to.

Real fisheries run on this: individual transferable quotas in New Zealand and Iceland since the 1980s. Alaska's halibut season had shrunk to a 2–3 day derby; under tradable quotas (1995) it runs about 8 months — same fish, no race.

Give the lake one owner.

The owner's profit *is* the net column — total catch minus boat costs. Reading it, the owner runs **4 boats** and pockets **\$1,600 a day**. No regulator, no table of fees.

Ownership makes the crowding cost land on the chooser.

A fifth boat would move \$400 between the owner's own decks and add \$300 of fish for \$400 of cost — **the owner simply declines**.

Own it, applied: elephants

Wildlife is a commons wherever it is **unowned**: a poached elephant is pure gain to the poacher; the loss falls on everyone — and on no one in particular.

- **Kenya**, 1973–1989: elephants unowned, hunting banned — the herd fell from roughly **167,000 to 16,000**. Bans without owners left no one whose loss a dead elephant was.
- **Southern Africa**: Zimbabwe's CAMPFIRE (from 1989) and its neighbors gave local communities rights to wildlife and its revenues — elephants became an asset to protect. The region now holds **the continent's largest herds**.

Where someone owns the elephant's value, someone guards it.

Own it together: Ostrom's commons

Between the market and the state: **communities that govern their own commons**
— and have, for centuries.

- Swiss alpine pastures: village grazing rules running since the Middle Ages.
- Maine lobster harbors: local territories and self-enforced limits.
- Spanish *huertas*: irrigation water shared by farmer-run courts for a thousand years.

Elinor Ostrom (*Governing the Commons*, 1990; Nobel 2009): what the survivors share
— **clear boundaries, rules written by the users, monitoring by the users, graduated penalties.**

The tragedy is a warning, not a prophecy.

The matrix, settled

		Rival in use?	
		yes	no
Excludable?	yes	Private goods a burrito	Club goods a streaming subscription
	no	Common resources fish in the open ocean	Public goods national defense

- **Private goods:** price is gate and meter — markets work.
- **Club goods:** gate holds; access sold by subscription.
- **Common resources:** overused — cap, price, permit, or own.
- **Public goods:** undersupplied — tax and provide, *if* benefits beat costs.

Active Learning 7

Pick the fix — and defend it. For each commons: which repair (cap / price / permit / own / community rules) fits best, and **what would its enforcer need to know?**

1. A village pasture shared by 30 families.
2. Downtown streets, gridlocked every evening.
3. A tuna fishery crossing a dozen national waters.
4. The campus gym at 6 p.m.

More than one defensible answer each — the match between fix and obstacle is the marks.

Four minutes. Paper and pencil.

Where no one can be kept out, the price system goes silent.

Public goods starve — no one can charge for supplying them.

Commons get devoured — no one is charged for using them up.

Every fix that works rebuilds the missing bill.